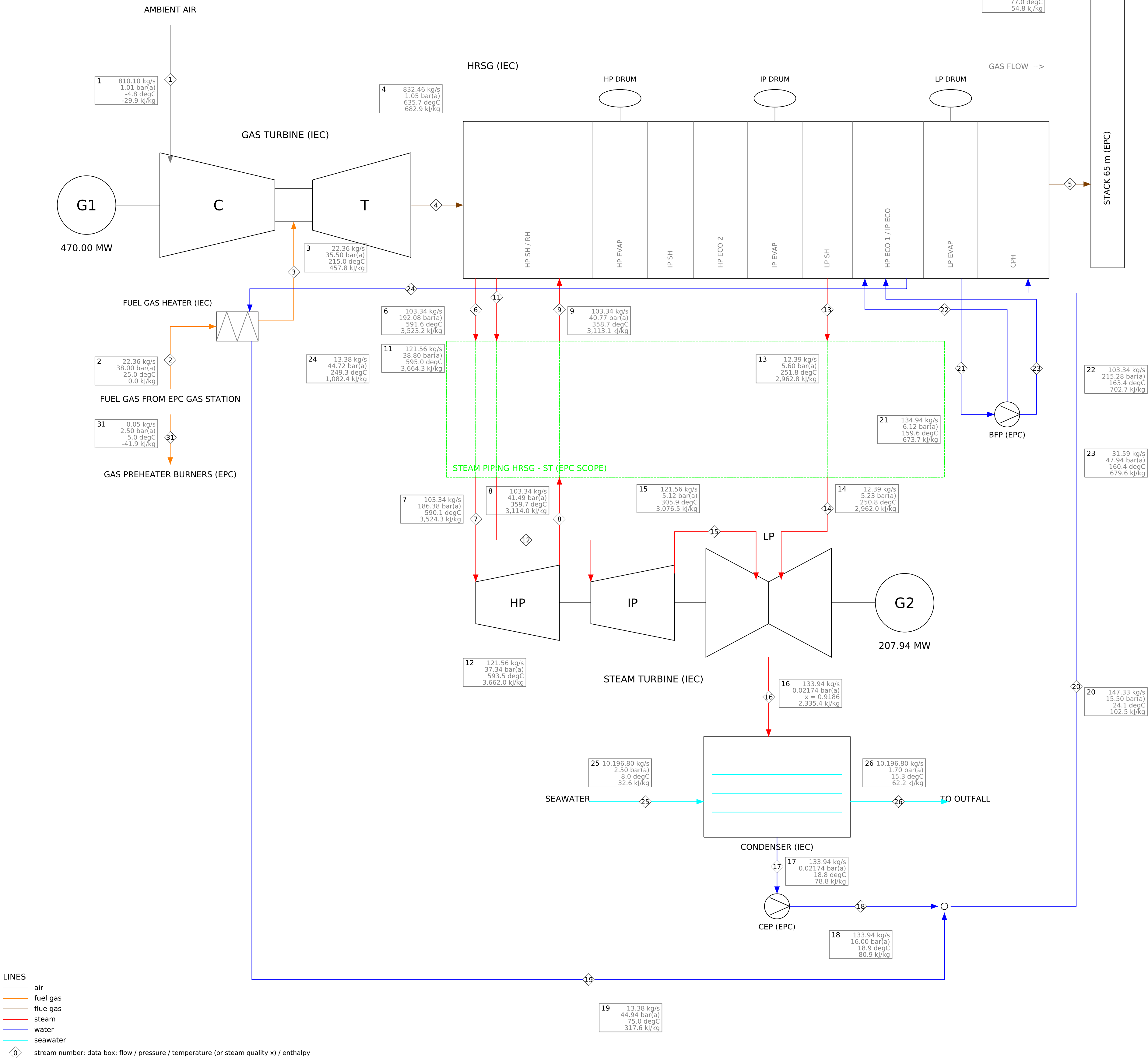




STREAM TABLE - WIN-NG-100

| No | Stream                                                              | kg/s      | bar(a) | degC / x | kJ/kg   | MW*   |
|----|---------------------------------------------------------------------|-----------|--------|----------|---------|-------|
| 1  | Ambient air to GT compressor inlet                                  | 810.10    | 1.01   | -4.8     | -29.9   | -24.2 |
| 2  | Fuel gas at performance gas heater inlet (IEC terminal point IF-01) | 22.36     | 38.00  | 25.0     | 0.0     | 0.0   |
| 3  | Heated fuel gas to GT combustion system                             | 22.36     | 35.50  | 215.0    | 457.8   | 10.2  |
| 4  | GT exhaust gas to HRSG inlet                                        | 832.46    | 1.05   | 635.7    | 682.9   | 568.5 |
| 5  | Flue gas at HRSG outlet to stack                                    | 832.46    | 1.01   | 77.0     | 54.8    | 45.6  |
| 6  | HP steam at HRSG outlet (HP SH outlet)                              | 103.34    | 192.08 | 591.6    | 3,523.2 | 364.1 |
| 7  | HP steam at ST main stop valve inlet                                | 103.34    | 186.38 | 590.1    | 3,524.3 | 364.2 |
| 8  | Cold reheat steam at HP turbine exhaust                             | 103.34    | 41.49  | 359.7    | 3,114.0 | 321.8 |
| 9  | Cold reheat steam at HRSG reheater inlet                            | 103.34    | 40.77  | 358.7    | 3,113.1 | 321.7 |
| 10 | IP superheated steam (joins cold reheat inside the HRSG)            | 18.21     | 41.62  | 330.4    | 3,039.2 | 55.3  |
| 11 | Hot reheat steam at HRSG outlet                                     | 121.56    | 38.80  | 595.0    | 3,664.3 | 445.4 |
| 12 | Hot reheat steam at ST IP stop/control valve inlet                  | 121.56    | 37.34  | 593.5    | 3,662.0 | 445.1 |
| 13 | LP steam at HRSG outlet                                             | 12.39     | 5.60   | 251.8    | 2,962.8 | 36.7  |
| 14 | LP steam at ST LP admission                                         | 12.39     | 5.23   | 250.8    | 2,962.0 | 36.7  |
| 15 | IP turbine exhaust (crossover, internal)                            | 121.56    | 5.12   | 305.9    | 3,076.5 | 374.0 |
| 16 | LP turbine exhaust to condenser (UEEP)                              | 133.94    | 0.0217 | x 0.919  | 2,335.4 | 312.8 |
| 17 | Condensate at hotwell outlet / CEP suction                          | 133.94    | 0.0217 | 18.8     | 78.8    | 10.6  |
| 18 | Condensate at CEP discharge                                         | 133.94    | 16.00  | 18.9     | 80.9    | 10.8  |
| 19 | Fuel gas heater water return to condensate                          | 13.38     | 44.94  | 75.0     | 317.6   | 4.3   |
| 20 | Condensate to HRSG condensate preheater inlet                       | 147.33    | 15.50  | 24.1     | 102.5   | 15.1  |
| 21 | Feedwater from LP drum to boiler feed pumps                         | 134.94    | 6.12   | 159.6    | 673.7   | 90.9  |
| 22 | HP feedwater at BFP discharge to HP economiser                      | 103.34    | 215.28 | 163.4    | 702.7   | 72.6  |
| 23 | IP feedwater at BFP interstage to IP economiser                     | 31.59     | 47.94  | 160.4    | 679.6   | 21.5  |
| 24 | IP economiser water to performance fuel gas heater                  | 13.38     | 44.72  | 249.3    | 1,082.4 | 14.5  |
| 25 | Circulating seawater to condenser inlet                             | 10,196.80 | 2.50   | 8.0      | 32.6    | 332.0 |
| 26 | Circulating seawater from condenser outlet                          | 10,196.80 | 1.70   | 15.3     | 62.2    | 634.0 |
| 31 | Fuel gas to the station preheater burners (water-bath heaters)      | 0.05      | 2.50   | 5.0      | -41.9   | -0.0  |

\* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

|                                       |         |        |
|---------------------------------------|---------|--------|
| GT output (generator terminals)       | 470.00  | MW     |
| ST output (generator terminals)       | 207.94  | MW     |
| Gross output                          | 677.95  | MW     |
| Auxiliary loads                       | -10.07  | MW     |
| Transformer losses (GSU, UAT, MV/LV)  | -2.17   | MW     |
| NET OUTPUT (HV terminals)             | 665.70  | MW     |
| Heat input (LHV)                      | 1,064   | MW     |
| Gross heat rate (LHV)                 | 5,637   | kJ/kWh |
| NET HEAT RATE (LHV)                   | 5,753   | kJ/kWh |
| Net efficiency (LHV)                  | 62.58   | %      |
| Fuel LHV (ISO 6976, from composition) | 47.48   | MJ/kg  |
| Fuel gas flow                         | 110,111 | Sm3/h  |
| Circulating water flow                | 36,055  | m3/h   |
| GT load                               | 100.00  | %      |
| PLANT LOAD (NET, % OF SRC-NG-100)     | 108.43  | %      |

CHECKS (guarantees, limits, balances)

|      |                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OK   | energy balance CMIX: residual -0.003 MW (tolerance 0.25 MW)                                                                                                                        |
| OK   | energy balance COND: residual +0.183 MW (tolerance 0.64 MW)                                                                                                                        |
| OK   | energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)                                                                                                                         |
| OK   | energy balance GT: residual +0.020 MW (tolerance 1.05 MW)                                                                                                                          |
| OK   | energy balance HRSG: residual +2.113 MW vs stated casing loss 2.091 MW (tolerance 1.00 MW)                                                                                         |
| OK   | energy balance PIPE: residual +0.260 MW (tolerance 1.17 MW)                                                                                                                        |
| OK   | energy balance ST: residual +0.000 MW (tolerance 0.85 MW)                                                                                                                          |
| WARN | CW temperature rise 7.28 K vs DP-CASE-0015 <= 7.0 K (Condenser cooling water temperature rise)                                                                                     |
| OK   | gas flow 110,111 Sm3/h = 91.8 % of DP-OWNR-0033 (120,000 Sm3/h)                                                                                                                    |
| OK   | net export 665.7 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1) |
| INFO |                                                                                                                                                                                    |

|          |                                                                                |
|----------|--------------------------------------------------------------------------------|
| PROJECT  | Alpilla 600 MW Combined Cycle Power Plant                                      |
| TITLE    | HEAT AND MASS BALANCE - WIN-NG-100                                             |
| CASE     | Winter design (-4.8 degC, 99.6 %), natural gas, base load (GT at output limit) |
| DOCUMENT | ALP-HMB-WIN-NG-100                                                             |
| STATUS   | preliminary - generated by engine hmb v1.2.0                                   |
| DATA     | 6b60fce553593bf5   2026-09-29                                                  |
| NOTE     | all values calculated from the database; do not edit                           |